Emerging Technologies: Spatial Computing and Quantum Computing

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Introduction

Software engineering is a rapidly evolving field that continues to be shaped by continuous advancements in computing technologies and the increasing complexity of modern software systems. Emerging trends in software engineering refer to new and evolving technologies, methodologies, and paradigms that significantly influence how software is designed, developed, deployed, and maintained. These trends typically arise in response to industry needs such as scalability, automation, intelligence, and improved user experience.

Recent studies show that the focus of software engineering research has been shifting in recent years toward data-driven and intelligent technologies. Areas such as artificial intelligence, cloud computing, and intelligent systems have become increasingly dominant, while some traditional research directions are gradually losing prominence over time due to changing industrial priorities and technological evolution [1]. This shift reflects a broader transformation in the field toward more adaptive, distributed, and computation-intensive systems.

Among the most important emerging technologies shaping the future of software engineering are spatial computing and quantum computing.

Spatial computing refers to technologies that merge digital content with the physical environment by enabling systems to understand and interact with space, location, and real-world context. It is already being adopted in applications such as augmented reality (AR), navigation systems, smart logistics, and geospatial analytics. For example, AR-based navigation tools demonstrate how digital instructions can be overlaid on real-world environments to improve user interaction. The main benefits of spatial computing include enhanced user experience, improved contextual awareness, and more intelligent decision-making in real time. However, challenges such as high computational requirements, data accuracy, and hardware limitations still need to be addressed for wider adoption [2].

Quantum computing, in contrast, introduces a fundamentally different computing paradigm based on the principles of quantum mechanics. It uses quantum bits (qubits), which can exist in multiple states simultaneously, enabling certain types of computations to be performed significantly faster than classical systems. Although still in the early stages of development, quantum computing is being actively explored in areas such as cryptography, complex optimization problems, drug discovery, and material simulation. Its main advantage lies in its potential to solve problems that are currently infeasible for classical computers. However, major challenges include hardware instability, error rates, and the difficulty of maintaining quantum states over time [3].

Overall, spatial computing and quantum computing represent two distinct but highly influential directions in the future of software engineering. Spatial computing enhances the integration between digital systems and the physical world, enabling more interactive and context-aware applications. Quantum computing, on the other hand, redefines computational power by introducing new ways of processing information that go beyond classical limitations. This report will explore both technologies in terms of their core concepts, real-world applications, benefits, challenges, and their expected impact on the future of software engineering.

1. Spatial Computing

1.1. Definition and Overview

Spatial computing names a family of approaches in which computation is tied to where things are, when they occur, and how people and machines move through space—so digital services are not limited to flat, context-free rectangles on a screen. In a broad, field-defining view, spatial computing spans the ideas, tools, technologies, and systems that create a shared understanding of locations: how we know them, communicate them, visualize relationships to them, and navigate through them [4]. A central motivation in that tradition is that knowing where you are in space and time can deepen understanding of neighbors, ecosystems, infrastructure, and the environment [4].

A consumer-facing usage of the same term—common in industry coverage—emphasizes mixed and augmented reality: anchoring graphics, UI, and agents to the real world using head-worn displays, passthrough video, depth sensing, and multimodal input (eyes, hands, voice, full-body motion) [5], [6]. In practice, many products blend geospatial "where on Earth" foundations (maps, coordinates, routing) with local "where in the room" tracking (SLAM, world-locked content, hand–eye calibration) [4], [5].

1.2. Current Adoption Status

Platforms and ecosystems. Public commentary often highlights Apple Vision Pro, Meta Quest family devices, and Microsoft offerings in mixed reality and cloud/geo services as visible anchors for investment and developer activity [5], [6].

Verticals. Adoption patterns include healthcare (surgical planning, training, telepresence), manufacturing (digital work instructions, remote expert assist), and gaming/entertainment—reinforced in explainer and analysis pieces aimed at business and technical readers [5], [6]. Separately, the GIS / location-intelligence thread remains pervasive in mobile maps, ride-hailing, emergency alerting, precision agriculture, and disaster response, reflecting spatial computing's long tail in everyday infrastructure [4].

Market shape. Many headsets remain premium or niche, but lower-cost VR, phone-based AR, and cloud spatial services broaden access compared with a decade ago [5], [6].

1.3. Potential Benefits and Challenges

Benefits. Organizations cite gains in 3D visualization for design and training, remote collaboration with shared spatial reference frames, and operational awareness when spatial analytics and live sensing align [4], [5], [6].

Technical challenges. Device-side constraints include battery, thermal limits, and compute budgets for real-time perception; distributed systems challenges include latency, map freshness, and sensor fusion across inconsistent conditions [4], [5].

Ethics and society. Continuous sensing (cameras, depth, gaze, microphones) intensifies privacy and surveillance questions—cousins of long-standing geoprivacy concerns in mobile location traces [4]. Organizational policies, consent, data minimization, and secure processing become part of the product, not an afterthought [4], [5].

1.4. Future Impact on Careers and Industries

Work may increasingly combine physical presence and shared virtual workspaces—not only screen sharing but copresent manipulation of models and data in 3D [5], [6]. Emerging or growing roles include XR/spatial software engineer, 3D interaction designer, spatial UX researcher, digital twin / simulation engineer, and geospatial data scientist—often requiring hybrid skills: real-time graphics, sensing, spatial reasoning, and responsible AI for perception pipelines [4], [5], [6].

2. Quantum Computing

2.1. Definition and overview

Quantum computing is a layered field that has roots in computer science, physics, and mathematics that utilize quantum mechanics to solve complex problems faster than on classical computers. [7]

The reason why these computers are faster than classical ones is because classical computers use binary bits to process operations, whereas quantum computers use qubits. Qubits are quantum bits that can exist in multiple states simultaneously, unlike binary bits (only 0 or 1 at a time). Quantum bits allow for parallel computation, which results in a faster computation for larger data.

Quantum computers are known for solving a certain type of complicated problem, utilizing quantum techniques such as superposition, entanglement, decoherence, and interference. These techniques are crucial to comprehend in order to fully understand quantum computing. Firstly, superposition states that tiny particles, qubits, can exist in multiple states simultaneously. Meaning you can add two or more quantum states, and the result will be another valid quantum state. This superposition enables the inherent parallelism in quantum computers. [7] Secondly, entanglement is a principle from quantum physics; it is a physical occurrence where qubits become so linked they become one, no matter the distance. Thirdly, interference is a foundational quantum mechanics, where qubits interact and influence themselves and other qubits while in a probabilistic superposition state [8]. Lastly, decoherence is the process in which a system in a quantum state collapses into a nonquantum state. Minimizing and avoiding decoherence is required for a good quantum computing system [3].

2.2. Current Adoption Status

Now that we've covered the theoretical principles that make quantum computing what it is today, it is important to dive into what quantum computing looks like now.

A modern quantum computer is a specialized machine, often housed in specialized, ultra-cold environments, connected to classical computers that manage the *qubit* operations. More recently, access to quantum computing has been facilitated by services such as Quantum-as-a-Service (QaaS), which is a cloud-based delivery model delivering remote access to quantum processors and tools.

Quantum science and technology are not new fields; the initial development of quantum machines and technologies has been around for nearly 100 years. Quantum computing was first proposed in the 1980s to improve physics simulation; it remained theoretical for a couple of decades until 1994, when Peter Shor created an altering algorithm that is considered one of the few known algorithms with compelling potential application [9].

Later on in 2001, IBM and Stanford University achieved a milestone by implementing Shor's Algorithm on a 7-bit computer, which was the first proof that quantum computers could perform calculations impossible for classical computers.

Today, the quantum computing industry is led by major technology companies such as Google, IBM, and Microsoft. Each one of these companies is pursuing a specific trend in practical applications of quantum computing. Google and IBM primarily utilize superconducting qubits built upon quantum dots (artificial atoms), whereas Microsoft focuses on topological qubits using Majorana particles. [10] IBM currently offers 20–30 quantum systems through the IBM Quantum Network and created Qiskit, an open-source software framework that allows developers to build and run quantum circuits. [10]

Moving forward, the most leading companies are targeting more commercially meaningful quantum advantages in specific practical applications such as drug discovery, material science, and optimization. In March 2025, a major achievement was done in a real-world application, running a medical device simulation on IonQ's 36-qubit computer that outperformed classical high-performance computing by 12 percent, one of the first documented cases of quantum computing delivering a practical advantage over classical methods. [11]

2.3. Potential Benefits and Challenges

Quantum computing has a lot of potential in a variety of industries, but this great potential is also met with a number of significant challenges. This section outlines the benefits and challenges in the realm of quantum computing.

The ability to tackle complex real-world problems is the number one benefit of this field, especially optimization problems on a scale that classical computers cannot do.

Furthermore, in logistics, specifically, quantum computing can optimize problems with countless variables, such as route planning for a global supply chain. In the financial sector, major companies have partnered with IBM to explore quantum algorithms for option pricing and risk analysis.

Cybersecurity also represents a critical benefit, although it can also be a disadvantage. An efficient modern quantum computer could break most of the encryption systems. nowadays, a cybersecurity threat known as "Q-Day." In 2024 the National Institute of Standards and Technology (NIST) has provided organization with encryption methods that can resist quantum attacks.

Despite these benefits and potentials, quantum computing has some serious technical and challenges. The most critical one is decoherence, as mentioned above. Qubits can easily lose their state due to a number of external factors, and it causes errors in computation that can greatly impact the computer’s effectiveness. Addressing this challenge requires quantum error correction code, which requires other qubits, resulting in computational overhead. Additionally, the number of professionals who are specializing in quantum computing, specifically quantum error correction, is only an estimated 1,800 to 2,000 professionals. This represents the talent challenge, which is the most pressing bottleneck alongside the technical challange.[12]

Lastly, it is important to note that quantum computing is not a general purpose replacement for classical computing. It only provides exponential benefits for particular kinds of problems, optimization, simulation, and cryptography, but classical computers are still superior for everyday computation.

2.4. Future Impact on Careers and Industries

Quantum Computing is expected to have an economic and industrial impact over the next decade. The quantum computing market, starting from an estimated $1 billion in 2024 and according to an extensive analysis done by McKinsey that states by 2035, quantum computing could be worth $28 billion to $72 billion [13]. This market research clearly shows that quantum computing has a huge impact on industries in the future. Various industries such as finance, logistics, pharmaceutical research are expected to benefit the most from this field.

In the case of software engineers, quantum computing is opening entirely different career paths. Common job titles for software-oriented quantum roles include quantum software developer, quantum algorithm engineer, quantum machine learning scientist, and quantum performance engineer.

In summary, the quantum computing field is expanding and transforming from a strictly theoretical research concept into a practical commercially relevant field.

Conclusion & Reflection

Working on this report really opened our eyes to just how quickly software engineering is evolving, and how unpredictable those changes can be. At first, spatial computing and quantum computing felt like separate worlds, just two shiny new ideas people were excited about. But the more we dug in, the more we noticed they're actually chasing a similar goal: pushing computers to see, represent, and calculate things in ways we haven't before, even if they're using totally different approaches.

One thing that stood out to us is how these two technologies feel like they're completing each other. Spatial computing is all about changing how we interact with machines by blending digital experiences into the real world. Meanwhile, quantum computing is rewriting the rules behind the scenes, changing how computers actually process information. Looking at both, it's clear that advances in software engineering aren't just about one path, they happen in how systems connect with people, and at the same time, in how those systems actually function underneath it all.

We also couldn't help noticing that both technologies come with their own mix of strengths and headaches. Spatial computing can create amazing user experiences and give apps a better sense of what's happening around them. But it's still wrestling with hardware limitations, privacy issues, and making sure everyone can access it. Quantum computing promises jaw-dropping capabilities, but right now, it's still stuck behind tricky challenges with qubits, noise, and some hardware that just isn't ready for prime time. It was a good reminder: new tech isn't just about what it might do someday, it's about what it can actually deliver, long-term.

From a software engineering perspective, this project made one thing really clear: the next generation of engineers will have to stay flexible. Stuff like: building for 3D spaces, designing XR experiences, inventing new algorithms, and keeping systems secure in a quantum future, they're all going to matter just as much as our current skills do right now. Plus, we can't put off thinking about privacy, fairness, and ethics until these technologies are mature. It has to start at the very beginning.

All in all, this project gave us a glimpse of where computing is heading. True innovation often shows up when we change the way people use technology, or overhaul the way computers work, or both. As future software engineers, if we start paying attention to these trends early on, we'll be ready to build systems that aren't just technically powerful, but also more people-centered, thoughtful, and trustworthy.

In conclusion, this paper has examined two of the most influential emerging trends in software engineering: Spatial Computing and Quantum Computing. Both trends clearly illustrate the ongoing shift from traditional computing approaches toward more advanced, intelligent, and immersive technologies that are redefining how software systems are designed, developed, and experienced. Spatial Computing, in particular, reflects a significant transformation in user interaction by blending digital elements with the physical environment, enabling more intuitive and engaging experiences. Its increasing adoption across industries such as healthcare, manufacturing, and gaming demonstrates its practical value in enhancing productivity, enabling richer remote collaboration, and improving training and education through 3D visualization. Despite these promising advantages, Spatial Computing still faces several technical challenges, including limited battery life, high processing requirements, and device comfort issues, in addition to human and ethical concerns such as privacy risks, motion sickness, and potential social isolation.

In parallel, Quantum Computing represents a groundbreaking evolution in computational capability by utilizing quantum mechanical principles such as superposition, entanglement, and interference. Unlike classical computing systems that rely on binary processing, quantum systems leverage qubits to perform parallel computations, allowing them to address highly complex problems more efficiently. Although the technology is still in its early stages and remains limited in terms of accessibility and scalability, its potential applications in areas such as cryptography, optimization, and scientific research are highly significant. However, challenges such as decoherence, high implementation costs, and the need for specialized expertise continue to limit its widespread adoption.

Overall, both Spatial Computing and Quantum Computing highlight the dynamic nature of software engineering as a field that continuously evolves to meet the demands of modern technology and society. These trends are expected to reshape industries, redefine workflows, and create new professional roles such as XR developers, spatial experience designers, and quantum engineers. As these technologies mature over the next decade, they will not only enhance system capabilities but also require software engineers to adopt interdisciplinary knowledge and continuously update their skills. Therefore, understanding and adapting to these emerging trends will be essential for future professionals who aim to remain competitive and contribute effectively to the advancement of the software engineering field.

Collaboration:

Student Name: Jory Alkhudari

Individual Contribution: In this assignment, I was responsible for writing the introduction section of the report. I explained what emerging trends in software engineering are in a simple and clear way. I also introduced the two main technologies covered in the report, which are spatial computing and quantum computing.

Student Name: Jana Ghazi AlHuthayli

Individual Contribution: In this assignment, I was responsible for researching and writing the Quantum Computing section of this report. It included explaining the concept and its foundation in relevant fields with the theoretical principles.Quantum principles were explained such as superposition, entanglement, interference, and decoherence. I also researched the history of quantum computing clarifying how it started and how we got here in terms of technologies. I also wrote about the current leading companies in this field and what real world applications are they developing. Additionally, I presented the benefits and challenges of quantum computing across industries. Then wrote an analysis of the future impact on careers, industries and for software engineers. Finally I discussed the potential career opportunities and paths for software engineers in the field of quantum computing. Most importantly all source I've used in writing my section was cited.

Student Name: Nora Mohammed Alkabkabi

Individual Contribution: In this assignment, I was responsible for writing the spatial computing section of the report. I explained the concept in a clear and simple way, including its definition, current adoption, benefits, challenges, and future impact. I focused on showing its importance as an emerging technology and how it can influence different industries and careers. I also supported my work with relevant references and reviewed the section to ensure clarity, organization, and accuracy.

Student Name: Arwa Alhassani

Individual Contribution: I wrote the Reflection section for our report. I read through the whole thing to make sure my thoughts actually built on our group's research instead of going off track.

I wanted to draw a clear connection between the two technologies. We showed how spatial computing is changing the way people interact with machines, while quantum computing is actually shifting how those machines process information behind the scenes. I also wrote about what these changes mean for future software engineers. Things like staying adaptable, and always thinking about ethics, privacy, and security from the beginning.

Besides writing that section, I joined our team's discussions and helped review everyone's work. I wanted to make sure everything read smoothly and fit together as a whole.

Student Name: Jana Saad Alotaibi

Individual Contribution: In this assignment, I was responsible for writing the conclusion section of the report. I summarized the main ideas that were discussed about spatial computing and quantum computing in a clear and simple way. I focused on explaining their importance as emerging technologies and how they can affect the future of software engineering and different industries. I also made sure the conclusion links all parts of the report together and gives a clear ending for the reader. In addition, I checked my work and made some small edits to improve the clarity and make sure the writing is easy to understand and organized.

References

[1] N. Medvidovic, “Software Engineering Research Trends 1994–2024: Stepping Beyond the Lamppost,” IEEE Transactions on Software Engineering, vol. 51, no. 3, pp. 685–?, Mar. 2025. Available: IEEE Xplore Full-Text PDF:

[2] S. Shekhar, S. K. Feiner, and W. G. Aref, “Spatial Computing: Knowing where you are in space and time promises a deeper understanding of neighbors, ecosystems, and the environment,” Communications of the ACM. Available:Spatial Computing – Communications of the ACM

[3] J. Schneider and I. Smalley, “What is Quantum Computing?,” IBM, 2024. Available: What Is Quantum Computing? | IBM

[4] PCMag, “What Is Spatial Computing? A Basic Explainer,” 2024. Available: https://me.pcmag.com/en/vr/17590/what-is-spatial-computing-a-basic-explainer

[5] C. Hackl, “What Is Spatial Computing?,” Forbes, Jan. 06, 2024. Available: https://www.forbes.com/sites/cathyhackl/2024/01/06/what-is-spatial-computing/

[6] Microsoft Research, “Spatial Computing,” 2022. Available: https://www.microsoft.com/en-us/research/wp-content/uploads/2022/07/2202.01493.pdf

[7] Amazon Web Services, “What is Quantum Computing?,” Amazon Web Services, Inc., Apr. 29, 2026. Available: https://aws.amazon.com/what-is/quantum-computing/ (accessed Apr. 30, 2026).

[8] G. Wright, “Quantum Interference,” WhatIs, 2023. Available: https://www.techtarget.com/whatis/definition/quantum-interference (accessed Apr. 30, 2026).

[9] “Shor’s Algorithm,” Wikipedia, Jan. 07, 2002. Available: https://en.wikipedia.org/wiki/Shor%27s_algorithm (accessed May 01, 2026).

[10] H. Agueny, “The Quantum Computing Race: IBM, Google & Microsoft,” IgniteHub, Mar. 12, 2025. Available: https://ignitehub.tech/insight/the-quantum-computing-race-ibm-google-and-microsoft/ (accessed May 01, 2026).

[11] “Quantum Computing Industry Trends 2025: A Year of Breakthrough Milestones and Commercial Transition,” SpinQ, 2025. Available: https://www.spinquanta.com/news-detail/quantum-computing-industry-trends-2025-breakthrough-milestones-commercial-transition (accessed May 01, 2026).

[12] M. U. Rehman, “Quantum Error Correction: Will Quantum Computers Overcome Their Biggest Challenge?,” The Quantum Insider, Mar. 16, 2026. Available: https://thequantuminsider.com/2026/03/16/understanding-quantum-error-correction/ (accessed May 01, 2026).

[13] H. Soller, “The Year of Quantum: From Concept to Reality in 2025,” McKinsey & Company, June 23, 2025. Available: https://www.mckinsey.com/capabilities/tech-and-ai/our-insights/the-year-of-quantum-from-concept-to-reality-in-2025 (accessed May 01, 2026).